

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 | SUB-STRAND 1.3 TEMPERATURE AND THERMAL EXPANSION | FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why did the sealed water bottle crack in the freezer, and what does this reveal about how all matter responds to changes in temperature? In this Final Explanation you tell the complete story of the cracked frozen bottle. A learner filled a clean 500 ml glass bottle to the brim with water, sealed it tightly with a metal cap and left it in a freezer overnight. By morning the bottle was cracked and a column of ice stuck out where the cap had been. Answer all five sections in the space under each prompt. Refer back to the cracked bottle throughout. Use scientific words accurately (temperature, kinetic energy, particles, thermal expansion, expansivity, density, lattice). If you can, you may also use the term hydrogen bonding, which you met in Lesson 6. Show every step of any calculation, with units. Draw and label clear particle diagrams where asked. Use examples from Kenya wherever you can. Write about 1,200 to 1,800 words in total. Write in your own words and use evidence from the experiments and graphs you worked with in Lessons 1 to 6.

Section 1: Explaining the Crack at the Particle Level

Look back at your Lesson 1 prediction and baseline model of the cracked bottle. (a) Draw and label two particle diagrams: liquid water just before freezing, and the ice formed in the bottle. (b) Using your diagrams, explain why the same mass of water takes up a different volume as ice than as liquid water. (c) Explain, step by step, how this led to the bottle cracking. Include the words sealed, pressure and tensile strength. (d) Most everyday materials get smaller when cooled. Explain why the cracked bottle surprised you in Lesson 1.

(a) Liquid water diagram: molecules close together, arranged randomly, moving and sliding past each other. Ice diagram: molecules fixed in an ordered hexagonal crystal lattice, vibrating about fixed positions, with larger gaps between neighbouring molecules than in the liquid. (b) In the liquid, water molecules sit closer together. When water freezes at 0 °C, the molecules arrange themselves into an open hexagonal lattice (in Lesson 6 we learned that hydrogen bonds hold them in this arrangement). The molecules are further apart on average, so the same mass takes up more space. Ice has a lower density (about 0.917 g/cm³) than liquid water (about 1.00 g/cm³), and so occupies a greater volume. (c) The bottle was filled to the brim and sealed, so the water had no room to expand. As the freezer cooled the water to 0 °C, it froze and its volume increased. The growing ice pushed outwards on the glass walls, so the pressure on the glass increased. When this pressure produced a stress larger than the tensile strength of the glass, the glass cracked along its weakest lines and the ice pushed out where the cap had been. (d) Our everyday experience, and most materials, tell us that things shrink when they get cold because the particles move less and come closer together. I predicted that the water would shrink. But water is anomalous: when it freezes it expands. The cracked bottle was clear evidence of this.

Section 2: Temperature and How We Measure It

(a) Explain what temperature tells us about the particles in a substance. (b) Water in the bottle began at about 18 °C and the freezer was at -18 °C. Convert both temperatures, and the freezing point of water, to kelvin. State the conversion you used. (c) Describe how a liquid-in-glass thermometer works and name its thermometric property. Then explain why the clinical thermometer you used at Station B would not be a suitable choice for measuring the freezer temperature. (d) Explain what happens to the kinetic energy of the water molecules as the freezer cools them, and link this to the cracked bottle.

(a) Temperature is a measure of the average kinetic energy of the particles in a substance. A higher temperature means the particles move or vibrate faster on average. (b) Use $T(K) = T(^{\circ}C) + 273$. Water at 18 °C: $18 + 273 = 291$ K. Freezer at -18 °C: $-18 + 273 = 255$ K. Freezing point of water: $0^{\circ}C + 273 = 273$ K. (c) A liquid-in-glass thermometer has a red alcohol column in a narrow glass tube. When the bulb is warmed, the alcohol expands and the column rises; when cooled it contracts and the column falls. Its thermometric property is the volume (length of the column) of the liquid, which changes continuously, noticeably and reproducibly with temperature. The clinical thermometer has a constriction that stops the column falling back before a reading is taken. This is designed for holding a body-temperature reading, so it is not suited to following a freezer temperature at -18 °C. A liquid-in-glass thermometer, whose column rises and falls freely, is the better choice for the freezer. (d) As the freezer removes thermal energy, the water molecules lose kinetic energy and move more slowly. At 0 °C (273 K) they slow enough to settle into the ordered hexagonal lattice. This lattice takes up more volume, and in the sealed bottle this caused the crack.

Section 3: Expansion of Solids and Quantifying It

(a) Explain at the particle level why a solid metal rod gets longer when heated. (b) Write the formula for linear expansion and say what each symbol means. (c) On the Standard Gauge Railway between Mombasa and Nairobi, a steel rail is 25 m long and its temperature rises by 45 °C. Using $\alpha = 11 \times 10^{-6}$ per °C, calculate the increase in length in millimetres. (d) A 10 m steel beam on a bridge in Kisumu experiences a 15 °C rise. Calculate its expansion in millimetres. (e) Explain why engineers leave gaps, and how this idea links to the cracked bottle.

(a) When a solid is heated, its particles gain kinetic energy and vibrate with greater amplitude. The average distance between neighbouring particles increases, so the whole rod becomes longer. (b) $\Delta L = \alpha L_0 \Delta T$, where ΔL is the change in length, α is the coefficient of linear expansivity of the material (per °C), L_0 is the original length and ΔT is the change in temperature. (c) $\Delta L = (11 \times 10^{-6})(25)(45) = 0.012375$ m, which is about 12.4 mm. (d) $\Delta L = (11 \times 10^{-6})(10)(15) = 0.00165$ m = 1.65 mm. (e) If a solid is not given room to expand, the expansion generates large forces against its boundaries and structures may buckle or crack. Engineers calculate ΔL using the formula and leave gaps, such as between SGR rails or at bridge joints, so the expansion can happen safely. In the cracked bottle the water had no space to expand and the confined expansion produced pressure that cracked the glass. The principle of expansion producing force against a rigid boundary is the same.

Section 4: Water Is Different: Evidence and Calculation

In Lesson 4 you cooled water in a flask with a capillary tube and plotted a temperature-volume graph. (a) Describe the shape of your graph as the water cooled from 20 °C to 0 °C and say what it shows about the volume at about 4 °C. (b) State the temperature at which water has its greatest density, and say how this differs from most substances. (c) Liquid water has a density of about 1.00 g/cm³ and ice at 0 °C about 0.917 g/cm³. Taking 500 ml of liquid water to have a mass of 500 g, calculate the volume of the ice and the increase in volume in ml, and give this as a percentage of 500 ml. (d) Explain why the formula $\Delta V = \gamma V_0 \Delta T$ does not by itself describe this freezing change.

(a) The graph is U-shaped. As water cooled from 20 °C the capillary column fell, so the volume decreased, until about 4 °C where the volume was at a minimum. Below 4 °C the column rose again, so the volume increased as the water cooled toward 0 °C. (b) Water has its greatest density at 4 °C, because its volume is smallest there. Most substances keep getting denser and smaller as they cool, so water is anomalous. (c) Volume of ice = mass ÷ density = 500 g ÷ 0.917 g/cm³ = about 545 cm³ (545 ml). Increase = 545 - 500 = about 45 ml. Percentage = 45 ÷ 500 × 100 = about 9%. (d) $\Delta V = \gamma V_0 \Delta T$ describes ordinary expansion where volume increases as temperature rises in a single phase. Here the volume increases as the temperature falls and the water changes state from liquid to a hexagonal ice lattice. This is a phase-change effect that goes beyond the formula. In the sealed bottle, a 9% increase of about 45 ml had nowhere to go, so the pressure on the glass rose until the glass cracked.

Section 5: Using Our Understanding in Kenya

(a) Describe one Kenyan example from Lesson 5 where thermal expansion is a problem engineers must design for, and one where it is used deliberately as a helpful tool. Name the expansion principle at work in each. (b) Kenya Power copper cables run between Nairobi and Thika. Explain why the cables are installed with some slack, and say what would happen on a cold morning if they were installed pulled tight. (c) Water pipes in cold highland areas

(a) Problem: expansion joints on the Thika Superhighway (and gaps between SGR rails) are included because the structure lengthens when heated and would buckle without room to expand; the principle is linear expansion, $\Delta L = \alpha L_0 \Delta T$. Helpful: a bimetallic strip in an electric iron thermostat. Two bonded metals expand by different amounts, so the strip bends and switches the heating off at the set temperature. Fitting a heated metal tyre onto a wooden cart wheel is another deliberate use: the tyre expands when hot, slips on, and tightens as it cools. (b) Cables lengthen when heated on hot afternoons, so they are installed with some slack so they can sag a little when they expand. If pulled tight, on cooler mornings the metal would contract, producing large tension that could snap the cable or damage the supports. (c) Water in a pipe that freezes expands by about 9% and, like water in the sealed bottle, has no space to expand. The pressure generated can exceed what the pipe can withstand, so the pipe bursts. (d) The cracked bottle reveals that changes in temperature change the spacing and motion of particles, so materials change size. Solids and liquids generally expand when heated and contract when cooled, and this can produce huge forces if the material is confined. Water is the anomalous case: it contracts when cooled to 4 °C, then expands as it freezes at 0 °C because its molecules form an open hexagonal lattice. Engineers calculate these changes, for example in expansion gaps, to turn a destructive force into a controlled tool.

such as Kericho and Nyandarua can burst. Use your answers above to explain why. (d) Return to the driving question. In your own words, what does the cracked bottle reveal about how matter responds to temperature changes?	
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Particle-level explanation of the cracked bottle	Clear labelled diagrams of liquid water and the hexagonal ice lattice. Explains greater spacing, larger volume, sealed container, outward pressure and tensile strength in a correct logical chain.	Explains that ice has more spacing and takes up more volume, and that the sealed bottle cracked because of pressure, but the chain of reasoning or the diagrams have small gaps.	States that water expands or that the bottle cracked but gives little or no particle-level reasoning, or diagrams are missing or incorrect.
Temperature, scales and thermometers	Defines temperature as average kinetic energy, converts all temperatures to kelvin correctly (291 K, 255 K, 273 K), and explains thermometric property and thermometer types accurately.	Mostly correct definition and conversions with a minor error or incomplete explanation of how thermometers work.	Definition of temperature is incorrect or confused with heat, or conversions and thermometer descriptions are mostly wrong or missing.
Calculations and use of evidence	All calculations (12.4 mm, 1.65 mm, 545 ml, 45 ml, 9%) are set out with formula, substitution, correct units and answers, and the temperature-volume graph is accurately described as evidence.	Most calculations are correct with steps shown; one slip in units or arithmetic, or the graph description is partly complete.	Calculations are incomplete or incorrect, steps are not shown, or evidence from the graph is not used.
Anomalous behaviour of water	Accurately describes the U-shaped graph, maximum density at 4 °C, the change in density from 1.00 to 0.917 g/cm ³ , and explains why $\Delta V = \gamma V_0 \Delta T$ does not by itself describe freezing.	Describes the anomaly and 4 °C correctly but gives limited explanation of density change or of the formula's limits.	Does not identify 4 °C or the anomaly clearly, or treats water like all other substances.
Kenyan applications and communication	Connects the bottle to at least two Kenyan examples (SGR, Thika Superhighway, Kenya Power cables, burst pipes in Kericho or Nyandarua) with correct principles, and answers the driving question clearly using accurate scientific vocabulary in organised writing.	Uses one or two Kenyan examples correctly, with a reasonable link to the driving question and mostly accurate vocabulary.	Examples are missing, vague or incorrect, the link to the cracked bottle is weak, and the writing is hard to follow.

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